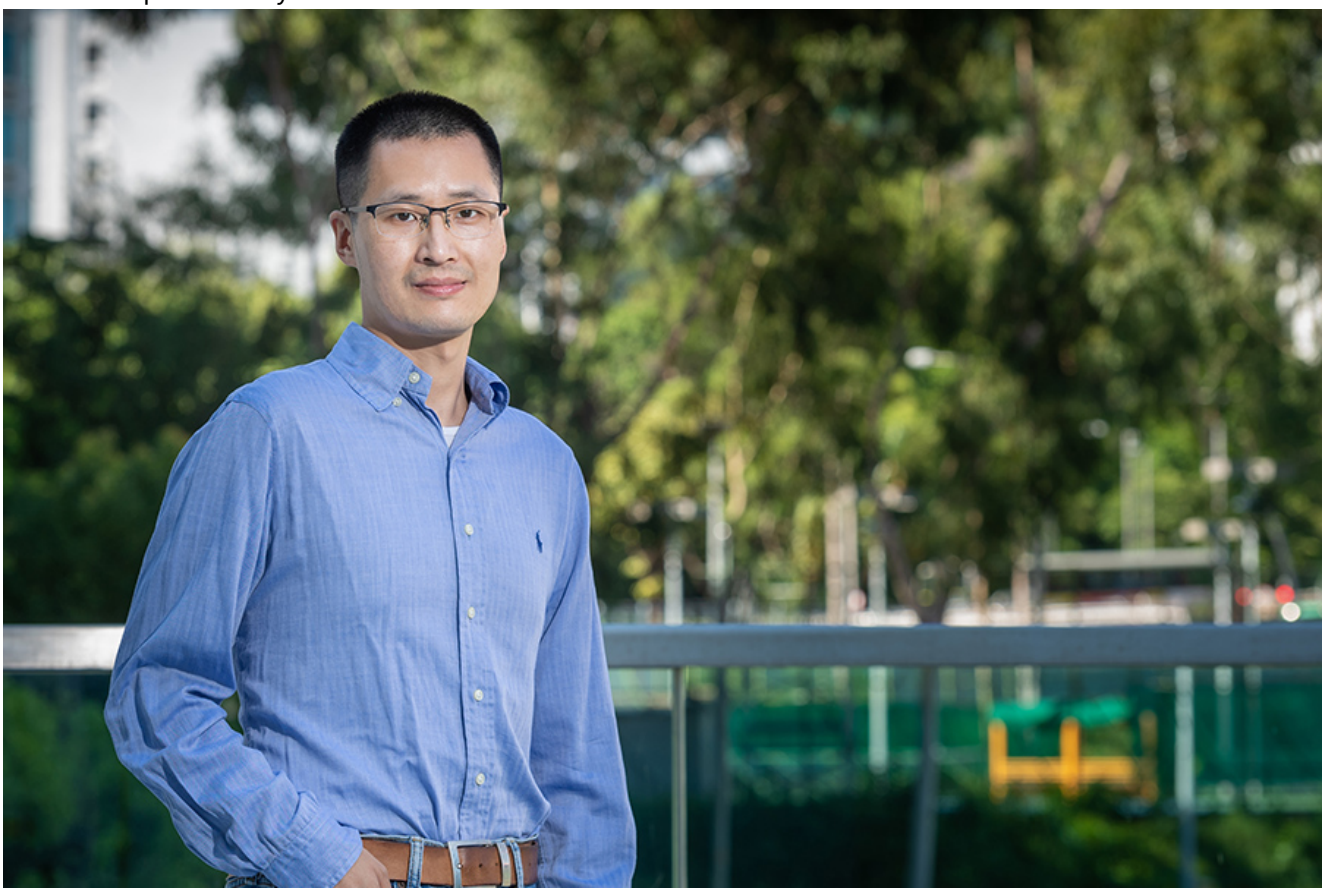

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About

Bo Han is currently an Associate Professor in Machine Learning and a Director of Trustworthy Machine Learning and Reasoning Group at Hong Kong Baptist University, and a BAIHO Visiting Scientist of Imperfect Information Learning Team at RIKEN Center for Advanced Intelligence Project (RIKEN AIP), hosted by Prof. Masashi Sugiyama, where his research focuses on machine learning, deep learning, foundation models, and their applications. He was a Visiting Research Scholar at MBZUAI MLD (2024), hosted by Prof. Kun Zhang, a Visiting Faculty Researcher at Microsoft Research (2022) and Alibaba DAMO Academy (2021), and a Postdoc Fellow at RIKEN AIP (2019-2020), working with Prof. Masashi Sugiyama. He received his Ph.D. degree in Computer Science from University of Technology Sydney (2015-2019), primarily advised by Prof. Ivor W. Tsang. He has co-authored three machine learning monographs, including Machine Learning with Noisy Labels (MIT Press), Trustworthy Machine Learning under Imperfect Data (Springer Nature), and Trustworthy Machine Learning from Data to Models (Foundations and Trends). He has served as Senior Area Chairs of NeurIPS and ICML, and Area Chairs of ICLR, UAI and AISTATS. He has also served as Associate Editors of IEEE TPAMI, MLJ and JAIR, and Editorial Board Members of JMLR and MLJ. He is an ACM Distinguished Speaker and an IEEE Senior Member. He received paper awards, including Outstanding Paper Award at NeurIPS, Most Influential Paper at NeurIPS, and Outstanding Student Paper Award at NeurIPS Workshop, and service awards, including Notable Area Chair at NeurIPS, Outstanding Area Chair at ICLR, and Outstanding Associate Editor at IEEE TNNLS. He received the RGC Early CAREER Scheme, NSFC AI Major Research Plan, IEEE AI's 10 to Watch Award, IJCAI Early Career Spotlight, INNS Aharon Katzir Young Investigator Award, IEEE Computing's Top 30 Early Career Professional Award, RIKEN BAIHO Award, Dean's Award for Outstanding Achievement, Microsoft Research StarTrack Scholars Program, and Faculty Research Awards from ByteDance, Baidu, Alibaba and Tencent.

Research Interests

- Foundation Models and Causal Representation Learning: [Monograph] [ICLR'24] [ICML'24] [NeurIPS'24] [ICLR'25] [ICML'25] [ICLR'26]
- Weakly Supervised and Self-supervised Representation Learning: [Survey] [NeurIPS'18] [ICML'19] [ICML'20] [ICML'21] [ICML'22] [ICLR'23]
- Robustness, Security and Privacy in Machine Learning: [NeurIPS'22] [ICLR'22] [NeurIPS'23] [NeurIPS'24] [ICLR'25] [ICLR'26]
- Federated, Efficient and Graph Machine Learning: [ICML'20] [KDD'21] [ICML'22] [NeurIPS'23] [ICLR'24] [ICLR'25]
- Interdisciplinary Problems (e.g., Healthcare Analytics and AI for Science): [NeurIPS'22] [Nature Plants'23] [ICLR'24] [NeurIPS'24] [ICLR'25]

TMLR Group is always looking for highly self-motivated PhD/RA/Visiting students and Postdoc researchers. Please read this document for recruiting information, and check this link and CSRankings for department information. Meanwhile, TMLR Group is happy to host remote research trainees. Due to the large number of emails we receive, we cannot respond to every email individually. Thanks!

Selected Publications

(see the updated full list here)

- Q. Wang, B. Han, Y. Liu, C. Gong, T. Liu, and J. Liu. W-DOE: Wasserstein Distribution-agnostic Outlier Exposure. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025.
- Z. Zhou, R. Tao, J. Zhu, Y. Luo, Z. Wang, and B. Han. Can Language Models Perform Robust Reasoning in Chain-of-thought Prompting with Noisy Rationales? In Advances in Neural Information Processing Systems 37 (NeurIPS'24), 2024.
- J. Yao, B. Han, Z. Zhou, Y. Zhang, and I.W. Tsang. Latent Class-Conditional Noise Model. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 45(8): 9964–9980, 2023.
- X. Xia, B. Han, N. Wang, J. Deng, J. Li, Y. Mao, and T. Liu. Learning with Mixed Open-set and Closed-set Noisy Labels. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 45(3): 3047–3058, 2022.
- A. Berthon, B. Han, G. Niu, T. Liu, and M. Sugiyama. Confidence Scores Make Instance-dependent Label-noise Learning Possible. In Proceedings of 38th International Conference on Machine Learning (ICML'21), 2021.
- B. Han, G. Niu, X. Yu, Q. Yao, M. Xu, I.W. Tsang, and M. Sugiyama. SIGUA: Forgetting May Make Learning with Noisy Labels More Robust. In Proceedings of 37th International Conference on Machine Learning (ICML'20), 2020.
- B. Han, Q. Yao, Y. Pan, I.W. Tsang, X. Xiao, Q. Yang, and M. Sugiyama. Millionaire: A Hint-guided Approach for Crowdsourcing. Machine Learning Journal (MLJ), 108(5): 831–858, 2019.
- B. Han, Q. Yao, X. Yu, G. Niu, M. Xu, W. Hu, I.W. Tsang, and M. Sugiyama. Co-teaching: Robust Training of Deep Neural Networks with Extremely Noisy Labels. In Advances in Neural Information Processing Systems 31 (NeurIPS'18), 2018.
- B. Han, J. Yao, G. Niu, M. Zhou, I.W. Tsang, Y. Zhang, and M. Sugiyama. Masking: A New Perspective of Noisy Supervision. In Advances in Neural Information Processing Systems 31 (NeurIPS'18), 2018.
- B. Han, Y. Pan, and I.W. Tsang. Robust Plackett-Luce Model for k-ary Crowdsourced Preferences. Machine Learning Journal (MLJ), 107(4): 675–702, 2018.